SYSTEM REQUIREMENTS FOR FLOORBALLKENYA.COM MVP

1. Functional Requirements

The FloorballKenya.com MVP shall provide the following core functionalities.

1.1 User Registration and Login

The system shall:

- Allow users to create accounts.
- Allow registered users to log in and log out.
- Allow users to update their profile information.
- Assign users different roles based on their responsibilities.

Restrict access to functions according to user roles.

The MVP will support the following user roles:

- Administrator
- Team Manager/Coach
- Player
- Vendor
- Customer

1.2 Team Management

The system shall allow authorized users to:

- Register a team.
- Create and update team profiles.
- Upload a team logo.
- Add players to a team.

- View team information.
- View team fixtures and results.

The team profile shall contain information such as:

- Team name
- Institution/club
- Team logo
- Coach/manager
- Contact information
- Registered players

1.3 Player Management

The system shall allow authorized users to:

- Register players.
- Create player profiles.

- Update player information.
- View player statistics.

The MVP shall maintain basic player statistics such as:

- Matches played
- Goals
- Assists
- Penalties

1.4 League Management

The administrator shall be able to:

- Create leagues.
- Create league seasons.
- Register teams into leagues.
- View participating teams.

- Manage league fixtures.
- Record match results.
- Display league standings.

The system shall automatically update the standings after a match result is recorded.
For example:
Win = 3 points
Draw = 1 point
Loss = 0 points
The system shall calculate:

- Matches played
- Wins
- Draws
- Losses
- Goals scored
- Goals conceded
- Goal difference
- Points

1.5 Fixture and Match Management

The system shall allow administrators to:

- Create fixtures.
- Assign teams to matches.
- Set match dates.
- Set match venues.
- Record match results.
- Update match information.

The system shall display:

- Upcoming matches
- Completed matches
- Match results
- Match scores

1.6 Tournament Registration

The MVP shall allow administrators to create tournaments and teams to register for them. The system shall:

- Display available tournaments.
- Allow teams to submit registration.
- Store tournament registration information.
- Allow administrators to approve or reject registrations.
- Display registered teams.

For the initial MVP, tournament registration can be implemented without complex automated tournament-bracket generation.

1.7 Marketplace

The MVP shall provide a basic multi-vendor floorball equipment marketplace.
Customers shall be able to:

- View products.
- Search for products.
- View product details.
- Add products to a shopping cart.
- Remove products from the cart.
- Place orders.
- View order status.

Products may include:

- Floorball sticks
- Floorball balls
- Jerseys
- Goalkeeper equipment

- Training equipment
- Floorball accessories

1.8 Vendor Management

The system shall allow vendors to:

- Register as vendors.
- Create vendor profiles.
- Submit vendor information for verification.
- Add products.
- Edit products.
- Update product prices.
- Update stock availability.
- View customer orders.

The administrator shall be able to:

- Review vendors.

- Approve vendors.
- Reject vendors.
- Suspend vendors.
- Manage vendor products.

1.9 Shopping Cart and Order Management

The system shall:

- Create a shopping cart for each customer.
- Store selected products.
- Calculate the total purchase amount.
- Generate an order number.
- Store order information.
- Allow customers to view their orders.

The MVP shall support basic order statuses:

Pending → Paid → Processing → Delivered →

Completed

1.10 M-Pesa Payment

The MVP shall provide M-Pesa payment functionality for marketplace purchases and, where implemented, tournament registration. The system shall:

- Initiate an M-Pesa payment request.
- Allow the customer to provide a phone number.
- Receive payment confirmation.
- Record the transaction.
- Link the payment to the relevant order.
- Update the order status after successful payment.
- Handle failed transactions.

kept simple. A full escrow and automated vendor payout system can be implemented in a later version.

1.11 Marketplace Commission

The MVP shall calculate the platform's commission from marketplace transactions.
For example:

Product price = KSh 5,000
Platform commission = 10%
Platform commission = KSh 500
Vendor amount = KSh 4,500

The administrator shall be able to:

- Set the commission percentage.
- View commissions generated.
- View vendor sales.

- View basic transaction reports.

1.12 Search

The system shall provide search functionality for:

- Players
- Teams
- Leagues
- Tournaments
- Products
- Vendors

1.13 News and Announcements

The administrator shall be able to publish basic floorball-related:

- News
- Announcements
- Tournament updates
- League updates

Users shall be able to view published content.

1.14 Administrator Dashboard

The system shall provide an administrator dashboard where the administrator can manage the platform.
The dashboard shall display basic information such as:

- Number of registered users

- Number of teams
- Number of players
- Number of leagues
- Number of tournaments
- Number of vendors
- Number of products
- Number of orders
- Total marketplace sales
- Platform commissions

The administrator shall also be able to manage users, teams, players, leagues, tournaments, products and orders.

1.15 Reports

The MVP shall generate basic reports including:
Sports reports

- League standings
- Match results
- Player statistics
- Team statistics

Marketplace reports

- Orders
- Product sales
- Vendor sales
- Platform commissions

Reports may be displayed on the administrator dashboard and exported where practical.

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2. Non-Functional Requirements

2.1 Usability

The system shall:

- Have a simple and user-friendly interface.
- Provide clear navigation.
- Use consistent page layouts.
- Provide understandable error messages.
- Be suitable for users with basic computer and smartphone skills.

2.2 Performance

The system shall:

- Load pages within a reasonable time under normal network conditions.
- Process database requests efficiently.

- Support multiple users accessing the platform simultaneously.
- Process normal marketplace transactions without unnecessary delays.

2.3 Security

The system shall:

- Require authentication for restricted areas.
- Implement role-based access control.
- Store passwords securely using appropriate password hashing.
- Validate user input.
- Protect administrative functions from unauthorized access.
- Use HTTPS.

- Protect M-Pesa API credentials.
- Maintain secure database access.

2.4 Reliability

The system shall:

- Maintain accurate player, team and match records.
- Prevent unauthorized modification of important records.
- Provide error handling for failed operations.
- Maintain regular database backups.
- Recover data where possible after system failure.

2.5 Scalability

The MVP shall be designed so that it can be expanded in the future to support:

- More teams
- More players
- More leagues
- More tournaments
- More vendors
- More products
- More users
- Other East African countries

2.6 Maintainability

The system shall:

- Use a structured database.
- Use modular system components.
- Follow consistent development practices.
- Maintain source code using version control.
- Provide documentation for important system components.
- Allow future developers to modify and extend the system.

2.7 Compatibility

The system shall be accessible using modern web browsers, including:

- Google Chrome
- Mozilla Firefox

- Microsoft Edge
- Safari

The interface shall be responsive and support:

- Smartphones
- Tablets
- Laptops
- Desktop computers

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3. Hardware Requirements

Client Requirements

Users shall require:

- Smartphone, tablet or computer
- At least 2 GB RAM

- Modern web browser
- Internet connection

Development Requirements

The developer shall require:

- Computer with at least 8 GB RAM
- At least 256 GB storage
- Modern processor
- Stable internet connection
- Code editor
- Web browser

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4. Software Requirements

The proposed MVP can use:

Component	Technology
CMS	WordPress
E-commerce	WooCommerce
Multi-vendor	Dokan/WCFM Marketplace
Database	MySQL/MariaDB
Payment	Safaricom M-Pesa Daraja API
Web Server	Apache/Nginx
Programming	PHP, HTML, CSS, JavaScript
Version Control	Git/GitHub
Browser	Chrome/Firefox/Edge

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5. Database Requirements

The MVP database shall maintain information for the following major entities:

- Users
- Roles
- Players
- Teams
- Leagues
- Seasons
- Tournaments
- Fixtures
- Matches
- Player statistics
- Team standings
- Vendors

- Products
- Categories
- Orders
- Payments
- Commissions
- News/Announcements

The database shall maintain appropriate relationships between these entities to ensure data consistency, integrity and minimal duplication.

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6. MVP Scope

To keep the project achievable within the university project period, the following features will form the core MVP:
Sports Management

- User management
- Team registration
- Player registration
- League management
- Fixture management
- Match results
- Automatic league standings
- Basic player statistics
- Tournament registration

Marketplace

- Vendor registration
- Product management
- Product browsing
- Shopping cart
- Orders
- M-Pesa payment
- Basic commission calculation

Administration

- Admin dashboard
- User management
- Sports management
- Marketplace management
- Basic reports

Features Excluded From the MVP

The following will be considered for future versions rather than the initial university implementation:

- AI chatbot
- Advanced scouting system
- Player transfer marketplace
- Automated social-media content
- Digital player cards
- Full escrow system
- Automated vendor payouts
- Advanced analytics

- East African regional expansion
- Complex sponsorship-management system